Short corn 2025

harvest data

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2025-11-04

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1 Rosemount

Date = 2025-10-31

Site = Rosemount, R22 and R34 Field

Methods = Harvested 43 ft of 2 rows at 30" row spacing in 6 adjacent corn rows. Moisture and test weight were estimated two ways…

1. using the HarvestMaster system on combine and
2. using a Perten 5200 on a subsample immediately afterwards

Does rosemount combine accurately determine moisture and test weight compared to benchtop equipment?

Data

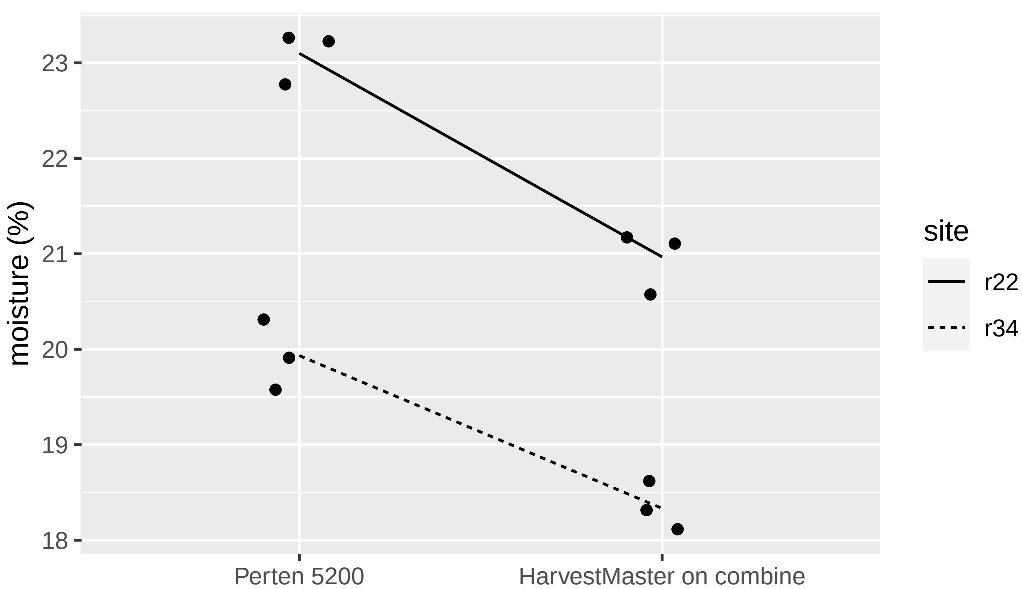
Testweight = good. with 0.5 lbs per bushel.

Moisture = ok, within 1.5 - 2 moisture %

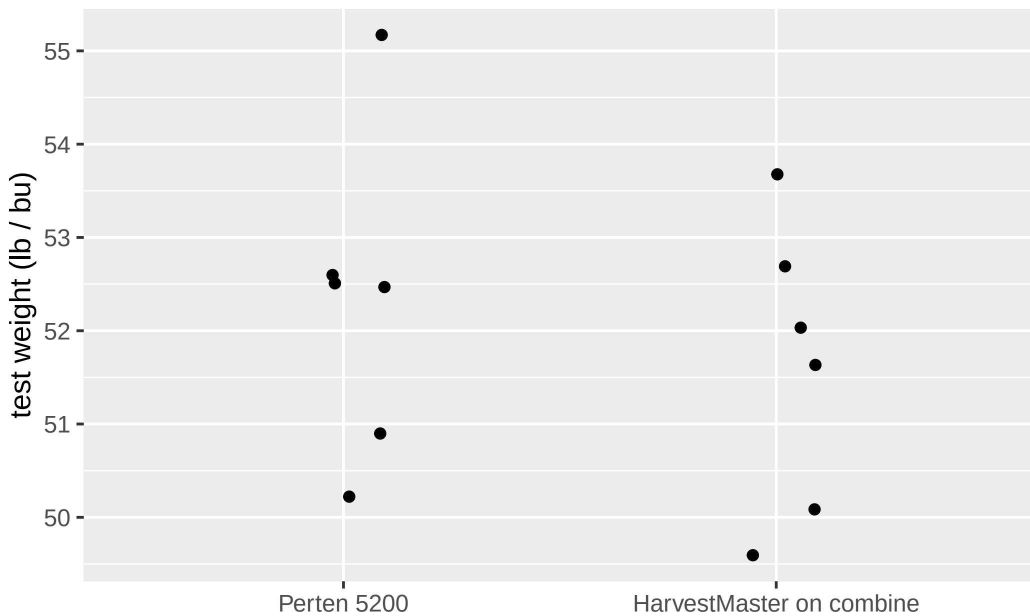
Discussion

The corn was bagged and driven back (1hr) in bed of truck during a light drizzle which could possibly have slightly increased moisture. When brought back to campus, they were put in dryer for 30 min, then analyzed.

1.1 moisture



1.2 test weight



1.3 data

1.3.1 raw

site	position	method	moisture	testweight
r22	middle	Perten 5200	22.8	52.5
r22	middle	HarvestMaster on combine	21.1	52.0
r22	north	Perten 5200	23.2	50.9
r22	north	HarvestMaster on combine	20.6	50.1
r22	south	Perten 5200	23.3	50.2
r22	south	HarvestMaster on combine	21.2	49.6
r34	east	Perten 5200	20.3	52.5
r34	east	HarvestMaster on combine	18.3	51.6
r34	middle	Perten 5200	19.9	52.6
r34	middle	HarvestMaster on combine	18.6	53.7
r34	west	Perten 5200	19.6	55.2
r34	west	HarvestMaster on combine	18.1	52.7

1.3.2 summarized

Table 2: moisture

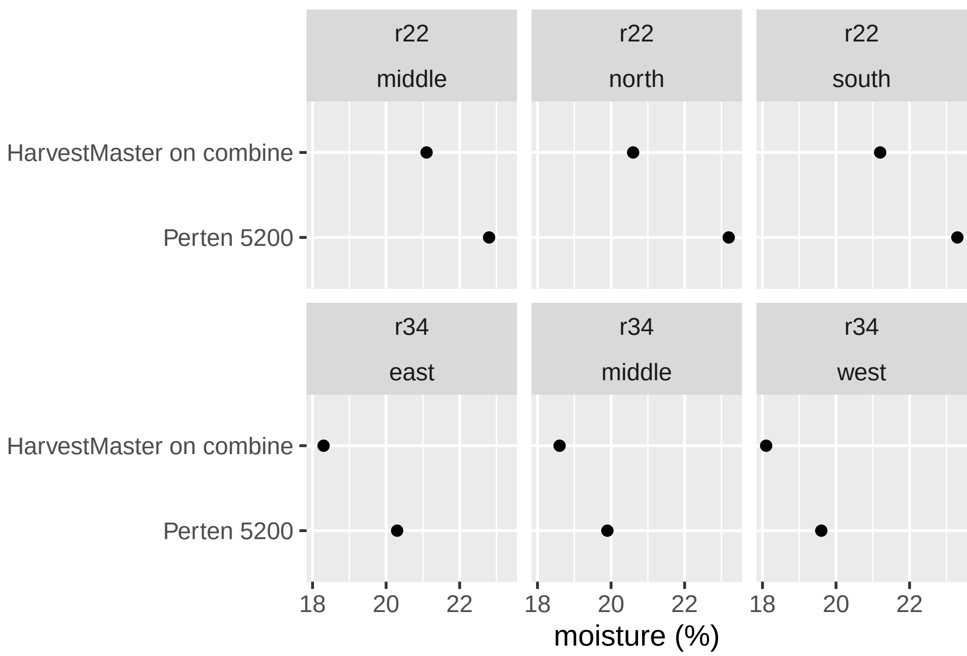
site	Perten 5200	HarvestMaster on combine	difference	%difference
r22	23.10000	20.96667	2.133333	9.235209
r34	19.93333	18.33333	1.600000	8.026756

Table 3: test weight

site	Perten 5200	HarvestMaster on combine	difference	%difference
r22	51.20000	50.56667	0.6333333	1.236979
r34	53.43333	52.66667	0.7666667	1.434810

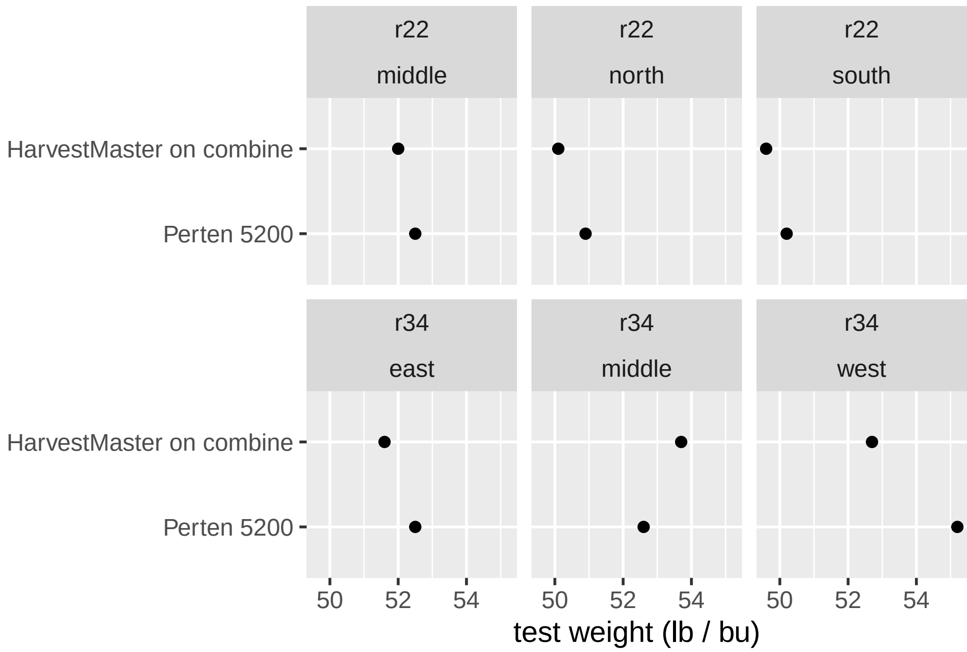
1.3.3 more in depth figures

1.3.3.1 moisture



- Trend is consistent across site*position

1.3.3.2 test weight



- Trend is consistent across site*position except R34 middle

2 St paul

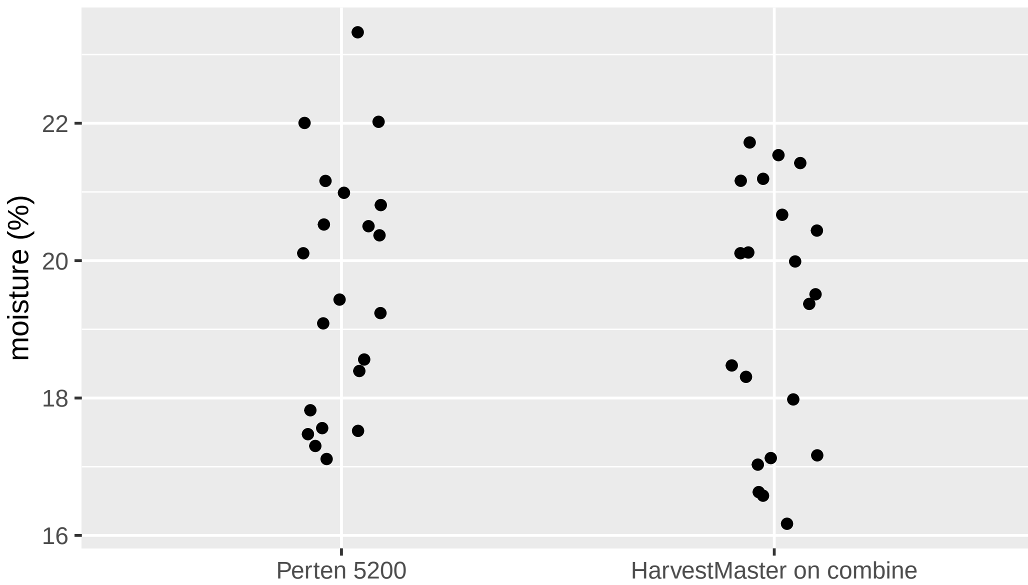
Date = 2025-11-3

Site = St Paul, C5

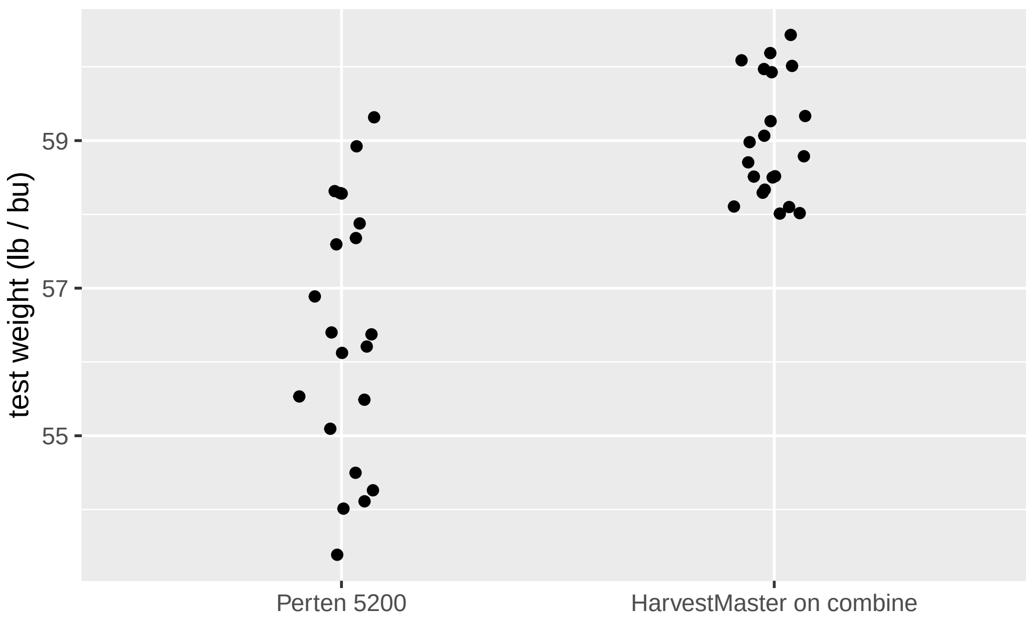
Methods = Harvested ~75ft of 2 rows at 30” row spacing. Moisture and test weight were estimated two ways…

1. using the HarvestMaster system on combine and
2. using a Perten 5200 on a subsample immediately afterwards

2.1 moisture



2.2 test weight

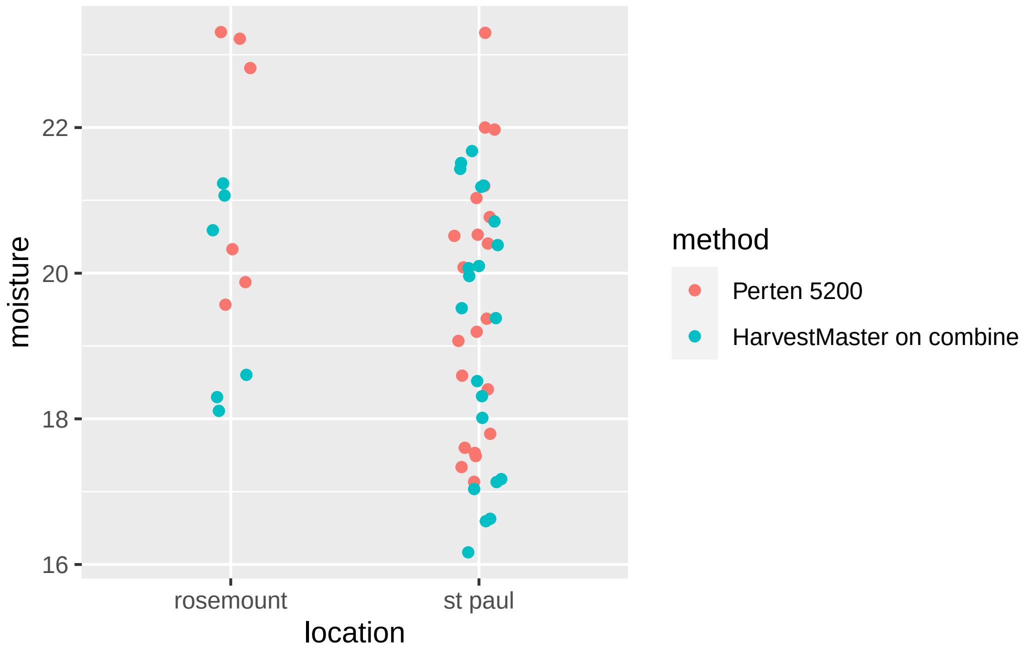


- harvestmaster on Thor's combine measured much less variability in test weight than the benchtop equipment

3 Both sites

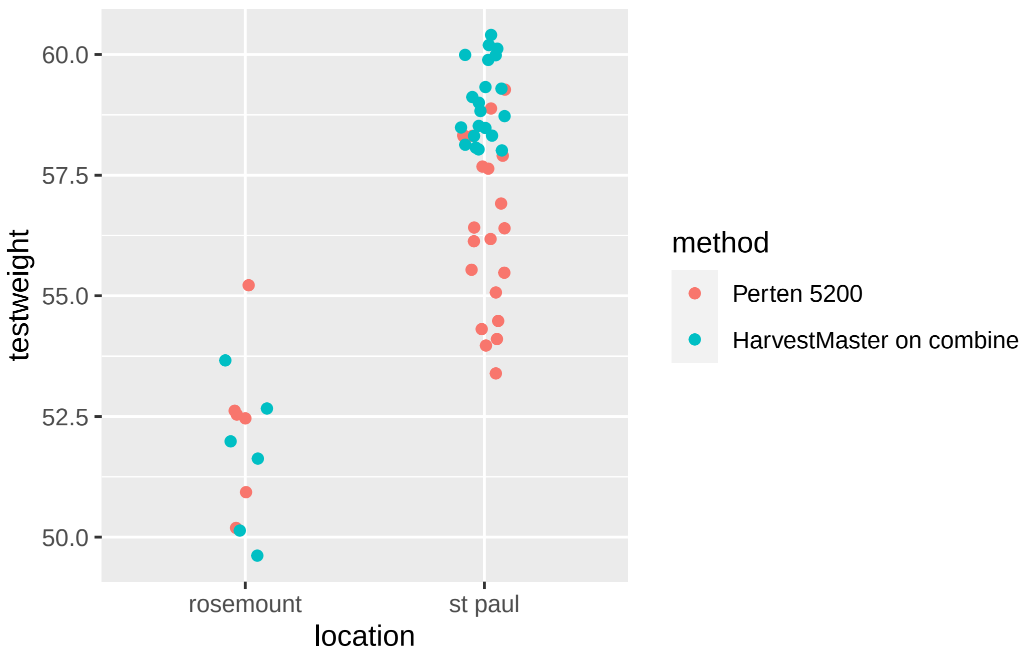
3.1 Yield (combine data only)

3.2 Moisture



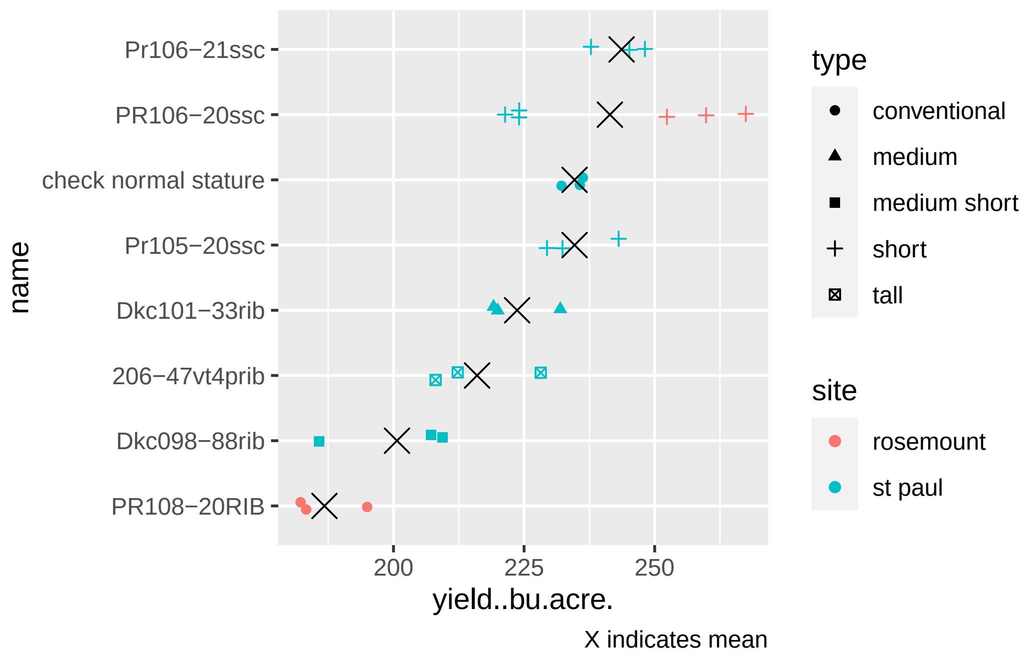
moisture
22
20
18
16
rosemount
st paul
location
method
Perten 5200
HarvestMaster on combine

3.3 Test weight



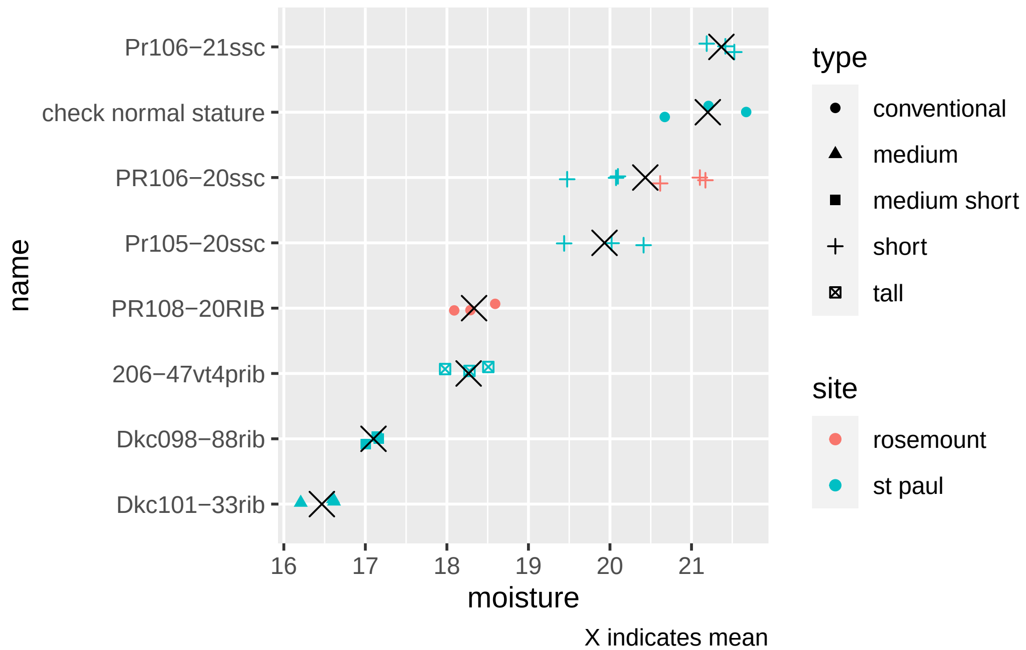
4 Variety performance (combine data only)

4.1 Yield



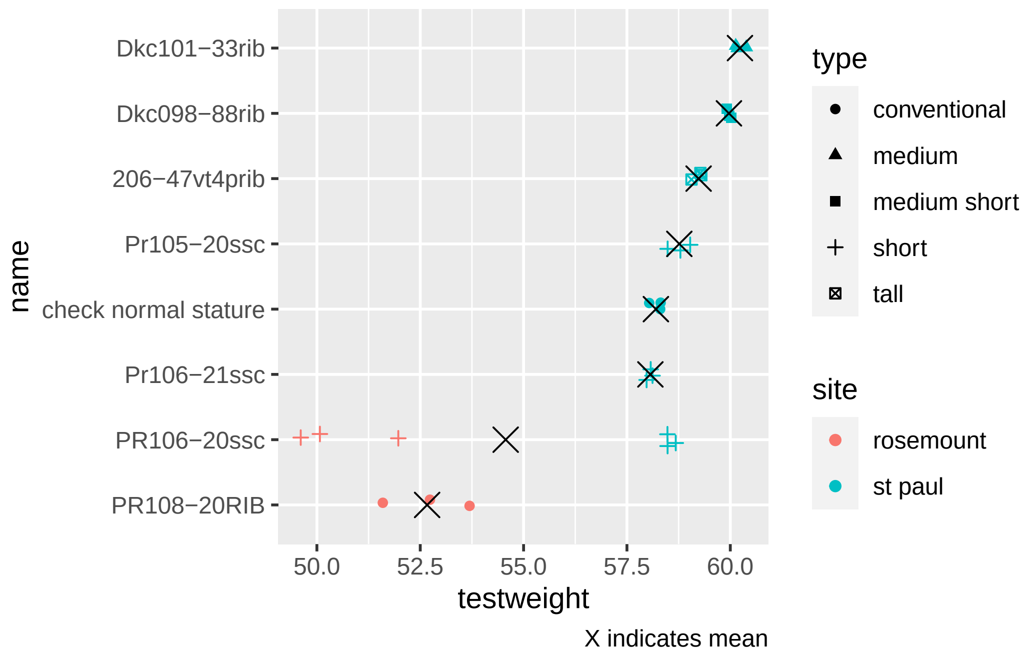
- Just shows combine harvest, which is new in 2025. Hand harvest estimates not included
- PR106-20ssc in rosemount was following alfalfa, may explain larger yield
- Need to add on 2 meters of row for each st paul strip and get precise lengths

4.2 Moisture



Pr106−21ssc
check normal stature
PR106−20ssc
Pr105−20ssc
PR108−20RIB
206−47vt4prib
Dkc098−88rib
Dkc101−33rib
name
16
17
18
19
20
21
moisture
X indicates mean
type
conventional
medium
medium short
short
tall
site
rosemount
st paul

4.3 Test weight



5 Photos

Figure 1: Short corn strip (2 rows at 43 feet long) harvested with wintersteiger quantum on 31Oct2025

Figure 2: Short corn strip (2 rows at 75 ft) harvested with Kincaid combine on 3Nov2025